Badee Khalbouss

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Education

University of Waterloo **2nd Year | Expected 2029**
Bachelor of Computer Science (Honours) *Waterloo, ON*

Experience

Penny.co **Jun 2025 – Sep 2026**
Full Stack Developer *Remote (Riyadh, Saudi Arabia)*

- Built an **AI assistant** that drafts RFQs/RFPs and routes them for approval, cutting manual request work by **50%**.
- Built an **in-house contract system** (drafting, e-signatures, tracking) replacing DocuSign in a **$150K+ contract**.
- Built **approval delegation** for a government commission's platform, supporting workflows for **1,000+ users**.
- Added **Arabic localization (i18n)** with right-to-left layouts, letting the platform serve the Saudi market.
- Made users' table **filters persist** across sign-ins and shareable by link using local storage and URL state.

First Robotics **Nov 2023 – Jun 2025**
Programming Team Head *Hamilton, ON*

- Led the team to **first place at the Niagara College championship**; wrote **Java** autonomous-drive code on a Raspberry Pi.

Career Education Council (CEC) **Jun 2024 – Aug 2024**
Student iOS Developer, {developthefuture} Virtual Co-op *Remote*

- Built an **AI iOS app** in **Swift** that uses **image recognition** to turn one meal photo into a per-item calorie breakdown.

Projects

Recital AI Agent | *Python, FastAPI, React, TypeScript, Ollama, MLflow* | GitHub | Demo **2026**

- Built a **RAG agent** for contract Q&A and risk review: **FastAPI** backend with background jobs, **React 19** frontend, local **Qwen 2.5 7B** via Ollama.
- Chunked contracts by clause with contextual headers; benchmarked **BM25**, **embedding** (bge-small) and **hybrid RRF** retrieval, shipping embeddings at **0.95 recall@5** and **0.80 MRR**.
- Added grounding checks (citation validation, sentence-level coverage, similarity gate), reaching **91% citation accuracy** and **100% correct refusals**.
- Added a verification pass to risk review, raising precision from **0.57 to 0.80** at full recall; logged every eval run in **MLflow**.

Co-op Tracker | *React, Ruby on Rails, JavaScript, MV3 Extension* | Live | GitHub | Demo **2026**

- Built a dashboard used by **100+ Waterloo students** to track co-op applications from WaterlooWorks and outside jobs.
- Built a sync pipeline (browser extension) that extracts **220+ applications in one click** and de-duplicates them by **Job ID**, so notes carry over across syncs.
- Kept student data private with browser-only storage on **GitHub Pages**, plus an optional **Rails 8 REST API**.

LEARN Deadline Tracker | *JavaScript, Chrome Extension (MV3)* | GitHub **2026**

- Built a **background crawler** that collects deadlines from LEARN, Odyssey and course pages into one place, with alarm-based reminders.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Ruby, Java, Swift, C
AI: RAG, hybrid retrieval (BM25, embeddings, RRF), Ollama, fastembed, MLflow, LLM evaluation, image recognition
Backend & Data: FastAPI, Node.js, NestJS, Express, Ruby on Rails, REST APIs, MongoDB, SQLite, Firebase
Frontend & Tools: React, Angular, NgRx, Vite, AG Grid, Git/GitHub, Nx, Linux, Chrome Extensions (MV3)